

Aditya Patadiya

aadityasoni901@gmail.com | +91 9409498914 | [LinkedIn](#) | [GitHub](#) | [Website](#)

Summary

- Software Engineer specializing in Python and Django, with experience building and integrating frontend components using React.
- Strong foundation in Linux, automation, and CI/CD practices, with exposure to containers and cloud environments.
- Open-source contributor with a strong focus on writing clean, scalable code and collaborating in fast-paced engineering teams.

Work Experience

Open-Source Contribution, [mitmproxy](#)

Rajkot, India | Dec '24 – Present

- Fixing Bugs and improving the tool with dedicated implementations and test cases.
- **PoW: PR's** [#7940](#), [#7483](#), [#7453](#)

Software Engineering Intern - Security & Platform Reliability at HUMAN INITIALS

Rajkot, India | June '25 – Sep '25

- Collaborated with engineering teams to improve platform reliability by analyzing system behaviour, writing internal tools, refining security workflows.
- Implemented logging, reports, and metrics dashboard to surface vulnerabilities, performance bottlenecks, and system health insights.
- Documented findings and proposed engineering solutions that improved code quality, secure design, and operations stability.

Skills

Programming Languages: Python, JavaScript, TypeScript, SQL, C, Node.js, Java

Frontend: React.js, TailwindCSS

Frameworks and Libraries: Django, Flask, Fast API, REST API, API Design, PyTest, Celery, WebSockets

Testing and Automation: PyTest, Unit Testing, Functional Testing

Databases: PostgreSQL, MySQL, MongoDB, SQLite

Cloud and DevOps: Docker, Docker Compose, Kubernetes, Azure, Github Actions, Jenkins, Git, Linux, Nginx, Redis

Other: Distributed Systems, Microservices, Integration Testing

Projects

Healthcare_Management_System, [GitHub](#)

Mar '25 – April '25

- **Django REST Framework • Next.js • PostgreSQL • Docker • PyTest • Redis**
- Engineered secure and scalable REST APIs with proper versioning, validation, and structured error handling using Django REST Framework.
- Developed automated unit and functional API tests and containerized services with Docker for stable multi-environment deployment.
- Implemented Redis-based background task processing to send user verification emails asynchronously after registration, improving response speed and user experience.
- Added logging, metrics and database performance monitoring to improve API reliability and response times.

AI-Driven Intrusion Prevention System with File Integrity Monitoring, [GitHub](#)

Oct '25 – Present

- **Python • Flask • REST API • Django • React.js • MySQL • ML • Docker • PyTest**
- Designing and Implementing a scalable, event-driven IPS combining FIM and real-time network anomaly detection with modular API architecture.
- Built automated unit & integration tests and added telemetry/logging to improve system reliability and detection accuracy.
- Integrating ML-based anomaly models with Explainable AI, reducing false positives and enabling stable, production-ready threat scoring.

Generative AI & Machine Learning Projects, [GitHub](#)

Nov '25 – Nov '25

- **Python • Langchain • Multi-Agent Systems • CI/CD • Azure**
- Developed multiple ML/GenAI systems using LangChain and multi-agent architectures focusing on scalable and modular design.
- Automated deployment using Azure-based CI/CD pipelines with integrated tests for reliable model updates.

PQC based Decentralized Secure Framework for Smart Parking System, [GitHub](#)

June'25 – Aug '25

- **Python • Flask • React.js • MongoDB • Blockchain • PQC (Kyber, SPHINCS+)**
- Developed distributed microservices for real-time parking allocation and secure booking.
- Implemented PQC-based encryption workflows ensuring high availability and fault tolerance.

ProvenanceGuard: Blockchain-Powered Autonomous Supply Chain, [GitHub](#)

Sep '25 – Present

- **Python • Django • Docker • SPADE (MAS) • Hyperledger Fabric**
- Building a decentralized supply chain architecture with Multi-Agent Systems that autonomously monitored and rerouted perishable goods, cutting waste by ~30%.
- Integrated Hyperledger Fabric for immutable audit logs and tested smart contract interactions.

Hybrid AES-GCM – Kyber Post Quantum Cryptography Algorithm, [GitHub](#)

Mar '25 – May '25

- **Python • Kyber • AES-GCM • SPHINCS+**
- Designed a modular enc-dec pipeline combining matrix-based operations with Kyber key exchange and SPHINCS+ signature.
- Structured the system as reusable, scalable components ready for integration into secure communication workflows.

Academic Details

Bachelor of Technology – Marwadi University (GPA: CGPA: 7.74)

Rajkot, India | Sep '22 – Present

Extra – Curricular Activity

Administrator | Cybersecurity Club | Marwadi University

Jan '25 – June '25

- Conduct 20+ sessions on Web Pentesting and mentored 60+ junior